

How to treat autism in children?

Evidence searched on 2026-10-08 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line For children with autism, evidence supports **exercise interventions** (improving motor skills) and **naturalistic developmental behavioral interventions (NDBIs)** (enhancing developmental and social outcomes). A **modified ketogenic diet combined with rehabilitation** shows short-term reductions in behavioral and symptom severity, but further validation is needed.

Most important recent finding Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills. [1]

Immediate context While behavioral interventions (e.g., NDBIs) and dietary approaches (e.g., modified ketogenic diet) show promise for specific outcomes, no single treatment is universally established for all autism-related symptoms. Motor skill improvements are distinct from broader developmental or behavioral benefits.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Aripiprazole	"Irritability Associated with Autistic Disorder"	DailyMed
Risperidone	"Treatment of irritability associated with autistic disorder"	DailyMed
NT1164 (latest, investigational)	Being tested in a registered Phase 3 trial, Not yet recruiting; not an approved treatment	NCT07257939

Latest findings

- **2026 Dec:** Observational study found altered sleep spindle characteristics in children with ASD, significantly associated with ASD diagnosis [2].
- **2026 Dec:** Randomized controlled trial showed a modified ketogenic diet combined with rehabilitation training significantly reduced ABC and CARS scores in young children with ASD after 2 months [3].
- **2026-09-22:** Case study reported sustained core symptom reduction in a 12-month follow-up of a child with severe ASD after bilateral dual-target deep brain stimulation [4].
- **2026-09-18:** Meta-analysis identified dose-response relationships between exercise interventions and improved overall, gross, and fine motor skills in children with ASD [1].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Behavioral interventions	Evidence-supported but limited	Behavioral interventions significantly improved social-emotional or challenging behavior outcomes in ASD [5].	[5]
Developmental interventions	Evidence-supported but limited	Developmental interventions improved social communication outcomes in early childhood ASD [5].	[5]

Approach	Evidence level	What the evidence shows	Reference
Exercise interventions	Emerging	Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response effects [1].	[1]
Modified ketogenic diet	Experimental or preliminary	A modified ketogenic diet combined with rehabilitation training reduced ABC and CARS scores in young children with ASD [3].	[3]
Deep brain stimulation (DBS)	Experimental or preliminary	Dual-target DBS led to sustained core symptom reduction in a 12-month case study of severe pediatric ASD [4].	[4]
Extended reality (XR) interventions	Emerging	XR interventions (e.g., VR, AR) showed promise in supporting children with ASD, particularly in training and educational systems [6].	[6]
Caregiver-assisted group-based NDBI	Emerging	Caregiver-assisted group-based Naturalistic Developmental Behavioral Interventions improved developmental and social outcomes in preschoolers with ASD [7].	[7]
Whole-plant cannabinoid extract	Experimental or preliminary	Disruptive behavior improved in 49% of children with ASD treated with whole-plant cannabinoid extract vs. 21% on placebo [8].	[8]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Aripiprazole	"Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder"	Route: Oral. Pediatric Patients (6 to 17 years) The recommended dosage range for the treatment of pediatric patients with irritability associated with autistic disorder is 5 to 15 mg/day. Dosing should be initiated at 2 mg/day. The dose should be increased to 5 mg/day, with subsequent increases to 10 or 15 mg/day if needed. Dose adjustments of up to 5 mg/day should occur gradually, at intervals of no less than one week.	FDA-approved for the use shown; label effective 2026-03-05; original approval date not retrieved	FDA label. DailyMed label

Drug	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	Route: Oral. Start with 0.25 mg per day for children weighing less than 20 kg, and 0.5 mg per day for those weighing 20 kg or more. After at least four days, increase the dose to 0.5 mg per day for children under 20 kg or 1.0 mg per day for those 20 kg or heavier, maintaining this dose for no less than 14 days. If insufficient response occurs, raise the dose by 0.25 mg per day for children under 20 kg or 0.5 mg per day for those 20 kg or more, with increases made every 2 weeks or more. The usual dose range is 0.5 mg to 3 mg per day, and this applies to children and adolescents weighing at least 15 kg.	FDA-approved for the use shown; label effective 2026-06-25; original approval date not retrieved	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed label

Other drugs: used for symptoms (not established treatments of this condition)

Off-label: a retrieved review or page describes use for specific symptoms or co-occurring conditions; the drug has no FDA-labelled indication for this condition. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Key evidence
quetiapine	irritability and aggressive behavior in children and adults with ASD (population: children and adults with ASD)	Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
ziprasidone	irritability and aggressive behavior in children and adults with ASD (population: children and adults with ASD)	Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
paliperidone	irritability and aggressive behavior in children and adults with ASD (population: children and adults with ASD)	Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
methylphenidate	ADHD-like symptoms	Recent Advances in Pharmacological Management of Autism: A Narrative Review
atomoxetine	ADHD-like symptoms	Recent Advances in Pharmacological Management of Autism: A Narrative Review

Recently approved or updated (FDA, EMA and CDSCO announcements)

Date	Type	Announcement	What the regulator's page says
date not stated	Regulatory announcement	FDA Responds to Evidence of Possible Association Between Autism and Acetaminophen Use During Pregnancy / FDA	Food and Drug Administration today initiated the process for a label change for acetaminophen (Tylenol and similar products) to reflect evidence suggesting that the use of acetaminophen by pregnant women may be associated with an increased risk of neurological conditions such as autism and ADHD in children. "The FDA is taking action to make parents and doctors aware of a considerable body of evidence about potential risks associated with acetaminophen," said FDA Commissioner Marty Makary, M.D., M.P.H.

Page wording from the regulator's own site; read the page for the full announcement.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
quetiapine , Lumateperone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS; and SUICIDAL THOUGHTS AND BEHAVIORS Increased Mortality in Elderly Patients with Dementia-Related Psychosis Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death [see Warnings and Precautions (5.1)]."
ziprasidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Ziprasidone is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)] ."
paliperidone , Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Paliperidone extended-release tablets are not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
methylphenidate	"ABUSE, MISUSE, AND ADDICTION Methylphenidate hydrochloride extended-release has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction."
atomoxetine	"SUICIDAL IDEATION IN CHILDREN AND ADOLESCENTS Atomoxetine increased the risk of suicidal ideation in short-term studies in children or adolescents with Attention-Deficit/Hyperactivity Disorder (ADHD). Anyone considering the use of atomoxetine in a child or adolescent must balance this risk with the clinical need. Co-morbidities occurring with ADHD may be associated with an increase in the risk of suicidal ideation and/or behavior."
Sertraline	"Suicidality and Antidepressant Drugs Antidepressants increased the risk compared to placebo of suicidal thinking and behavior (suicidality) in children, adolescents, and young adults in short-term studies of major depressive disorder (MDD) and other psychiatric disorders. Anyone considering the use of Sertraline hydrochloride or any other antidepressant in a child, adolescent, or young adult must balance this risk with the clinical need."
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."

These are the labels' own boxed warnings, not a complete safety review: read the full label.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07863752	Stronger Together Mindfulness-Based Intervention; Psychoeducation Program	Phase not applicable	Completed (2026-10-07)	Completed; results not posted/retrieved.
NCT07076264	AquOTic	Phase not applicable	Recruiting (2026-10-07)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06659328	Structured Walking Intervention for Children with Autism Spectrum Disorder.	Phase not applicable	Completed (2026-10-06)	Completed; results not posted/retrieved.

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.
NCT07257939	NTI164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07243561	hUC-MSC-Exos Nasal Spray	Phase 1/2	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03279471	Sertraline; CBT/BIACA	Phase 2	Completed (2026-09-23)	Completed; results posted (outcome not retrieved here).
NCT07424404	KarXT; KarX-EC	Phase 3	Recruiting (2026-09-22)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04630847	Fecal Microbiota Transplant	Phase 1	Not yet recruiting (2026-09-18)	Under investigation; results not yet available; effectiveness cannot be inferred.

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Sandbank et al. (2023)**, Meta-analysis: Behavioral and developmental interventions showed significant effects on social-emotional and communication outcomes in early childhood ASD [5].
2. **Wang et al. (2026)**, Meta-analysis: Exercise interventions demonstrated dose-response improvements in motor skills for children with ASD [1].
3. **Liu et al. (2026)**, Randomized controlled trial: A modified ketogenic diet combined with rehabilitation training reduced ASD symptom severity scores (ABC, CARS) in young children [3].
4. **Seng et al. (2026)**, Randomized controlled trial: Caregiver-assisted group-based NDBI improved developmental and social outcomes in preschoolers with ASD [7].

Conflicting evidence

None was retrieved.

What remains under investigation

- **Efficacy:** Long-term effects of deep brain stimulation (DBS) for severe ASD symptoms beyond 12 months.
- **Effectiveness:** Real-world effectiveness of caregiver-assisted group-based NDBI in diverse clinical settings.
- **Feasibility:** Scalability of modified ketogenic diet interventions in routine pediatric care.
- **Association:** Neurophysiological markers (e.g., sleep spindles) as predictors of ASD symptom progression.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 2 retrieved record(s) were excluded because they were not about the question's subject.
- The PubMed / Europe PMC search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

Sources

1. Wang T, Dong X, Jia Y, et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. [PMID 42758334](#). DOI 10.1007/s00431-026-07421-4.
2. Erçelebi H, Özbudak P, Menderes D, et al. Altered sleep spindle characteristics in children with autism spectrum Disorder: A potential neurophysiological marker. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42753649](#). DOI 10.1016/j.clinph.2026.2112405.
3. Liu L, Zhao Q, Zhou J, et al. Evaluating the therapeutic efficacy of a modified ketogenic diet in children with autism spectrum disorder: a randomized controlled trial. *BMC pediatrics*. 2026-09-11. [PMID 42763398](#). DOI 10.1186/s12887-026-07708-3.
4. Shi M, Hu J, Pei J, et al. Dual-target deep brain stimulation for severe pediatric autism: Sustained core symptom reduction in a 12-month case study. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42759099](#). DOI 10.1016/j.clinph.2026.2112403.
5. Sandbank M, Bottema-Beutel K, Crowley LaPoint S, et al. Autism intervention meta-analysis of early childhood studies (Project AIM): updated systematic review and secondary analysis. *BMJ (Clinical research ed.)*. 2023 Nov 14. [PMID 37963634](#). DOI 10.1136/bmj-2023-076733.
6. Hao D, Mansi H, Elkefi S, et al. Extended reality (XR) interventions in supporting children with autism: A systematic review. *International journal of medical informatics*. 2026 Dec. [PMID 42784985](#). DOI 10.1016/j.ijmedinf.2026.106719.
7. Seng GJ, Lin JY, Huang WL, et al. Feasibility and Preliminary Efficacy of a Caregiver-Assisted Group-Based Naturalistic Developmental Behavioral Intervention Program for Autistic Preschoolers: A Pilot Randomized Controlled Trial. *Autism : the international journal of research and practice*. 2026-07-23. [PMID 42494078](#). DOI 10.1177/13623613261462574.
8. Aran A, Harel M, Cassuto H, et al. Cannabinoid treatment for autism: a proof-of-concept randomized trial. *Molecular autism*. 2021 Feb 3. [PMID 33536055](#). DOI 10.1186/s13229-021-00420-2.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- [Pharmacological treatments in autism spectrum disorder: a narrative review | Journal of Psychopathology](#)
 - [Recent Advances in Pharmacological Management of Autism: A Narrative Review](#)
 - [FDA Responds to Evidence of Possible Association Between Autism and Acetaminophen Use During Pregnancy | FDA](#)
 - [FDA drug label: Aripiprazole \(label effective 2026-03-05\)](#)
 - [FDA drug label: Risperidone \(label effective 2026-06-25\)](#)
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Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.